Momentum in Levels, Reversion in Spreads: A Systematic Study of U.S. Treasury Futures Curve Strategies, 2010–2026

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***Abstract*—We evaluate systematic momentum and curve mean-reversion strategies on U.S. Treasury futures over January 2010 to July 2026 using Bloomberg data. Continuous, back-adjusted futures are reconstructed from the exchange roll map, and curve trades are hedged to be DV01-neutral. Our central finding is a clear asymmetry between levels and spreads: outright duration exhibits trend, so a time-series momentum strategy earns a net Sharpe ratio of 0.68, whereas the yield-curve slope exhibits mean-reversion. A naive momentum signal applied to the 2s10s slope is empirically duration-neutral yet loses money (net Sharpe -0.79), while the same book driven by a mean-reversion signal earns 0.71. Two curve mean-reversion strategies (2s10s and 5s30s) perform comparably to the outright-momentum baseline but do not statistically dominate it, and a 2s5s10s butterfly does not earn its risk. All results are net of transaction costs and scaled to a common volatility target. We discuss the methodological requirements—roll reconstruction, DV01 hedging, and leakage-free backtesting—that materially affect these conclusions, and we are explicit about the out-of-sample validation still required before the curve results can be treated as a hard performance claim.**

***Index Terms*—fixed income, Treasury futures, time-series momentum, mean-reversion, yield curve, systematic strategies, backtesting.**

I. Introduction

Systematic strategies in fixed income are commonly organized around three premia: carry, momentum, and value/mean-reversion. Time-series momentum has been documented across asset classes [1], and carry has been shown to be a broadly compensated factor [2]. Fixed-income practitioners have long distinguished *outright* directional exposure (the level of rates) from *relative-value* curve exposure (the slope and curvature between maturities) [3]. These two exposures need not share the same time-series behavior, and conflating them—for example, applying a single momentum overlay to both—can be actively harmful.

This paper makes that distinction concrete on the most liquid expression of the U.S. rates market, CME Treasury futures. We study the momentum and mean-reversion premia only; carry is outside the scope of this paper. We construct a reproducible pipeline that (i) rebuilds continuous futures from the exchange roll map, (ii) hedges curve trades to DV01 neutrality, and (iii) backtests a small, pre-specified set of strategies under a leakage-free protocol. Our contribution is not a new signal but a clean, honestly-reported empirical contrast: over 2010–2026, outright duration trends while the curve mean-reverts, and we quantify the cost of getting that distinction wrong.

II. Data

All series are daily Bloomberg exports covering January 2010 to July 2026, inner-joined onto the trading calendar of the front 10-year note future, yielding approximately 4,160 observations. The dataset comprises: the constant-maturity cash Treasury curve (3M, 6M, 1Y, 2Y, 3Y, 5Y, 7Y, 10Y, 20Y, 30Y); the SOFR/OIS and (legacy) swap curves together with Fed funds futures and the policy target rate for the macro overlay; and the futures complex itself across six roots—2Y (ZT), 5Y (ZF), 10Y (ZN), Ultra 10Y (TN), classic Bond (ZB), and Ultra Bond (UB)—each with the first three generic contracts and per-contract static specifications. The MOVE index, VIX, and 10-year breakeven inflation complete the volatility and macro controls.

Three data-handling issues materially affect correctness and are addressed explicitly in the pipeline. First, Bloomberg stamps prior values on non-trading days in the cash index series; the inner join to the futures calendar removes these phantom rows. Second, contract-level date fields are returned as locale-dependent text, which can silently transpose day and month; these are parsed day-first and validated against the futures month code. Third, and most importantly for return construction, the mapping from each generic to its underlying contract is retained so that rolls can be identified and neutralized rather than contaminating returns.

III. Methodology

A. Continuous futures

For each root we take the front generic's settlement series and identify roll dates as changes in the underlying-contract identifier. We then apply a difference back-adjustment: the settlement gap introduced on each roll date is treated as a contract switch rather than a price move, and the cumulative sum of all *future* roll gaps is subtracted from each historical level. The resulting series has no artificial roll jumps, and daily returns are computed from the adjusted level.

B. DV01-neutral hedging

Curve trades require the two (or three) legs to be balanced in interest-rate sensitivity. We use each contract's daily forward DV01 as published by Bloomberg (the risk to delivery of the cheapest-to-deliver bond, adjusted by its conversion factor), taken as a time series over the full sample. The hedge ratio for a spread is the ratio of the two legs' DV01s, lagged one day. This holds the realized books duration-neutral: the regression of the 2s10s book's daily P&L on the 10-year yield change has a slope of -0.001, indistinguishable from zero. Using the contract's own forward risk avoids the estimation noise and lag of a regression-based DV01 proxy, particularly around delivery and cheapest-to-deliver switches.

C. Signals

Two signal families are used, each matched to the exposure it drives:

- **Outright duration (momentum).** The sign of the trailing 63-day return of each root's continuous series.
- **Curve (mean-reversion).** The negative standardized deviation of the relevant spread (2s10s, 5s30s) or curvature (2s5s10s) from its own trailing 252-day mean, clipped at ± 3 and lagged one day.

The choice of mean-reversion for the curve is motivated economically—slopes and spreads are stationary relative to levels—and is examined empirically in Section IV.

D. Backtest protocol

Positions are formed from information available at $t-1$ and executed at the t settlement. Rebalancing is monthly. Each book is scaled toward a 10% annualized volatility target using 63-day realized volatility, with leverage capped at $3\times$ and a 126-day warm-up before any position is taken, so that a small early-sample volatility estimate cannot inflate the book. Transaction costs are charged at one tick per contract per side, using the per-contract tick size from the static specifications (the 10-year tick, for example, was halved over the sample, which the per-contract data captures).

E. Evaluation

We report the annualized Sharpe ratio (net of costs), maximum drawdown, and annual turnover. To compare each strategy against the outright-momentum baseline we use the Diebold–Mariano test [4] on the daily net-return differential. Because positions are rebalanced monthly, a holding is carried for roughly twenty-one trading days and the daily return differentials are strongly autocorrelated; we therefore estimate the long-run variance with a Newey–West (Bartlett-kernel) HAC estimator at a lag spanning the monthly holding, and apply the Harvey–Leybourne–Newbold small-sample correction with Student-t critical values. A positive statistic means the alternative out-returns the baseline. We verified the implementation against a HAC-robust regression of the differential on a constant.

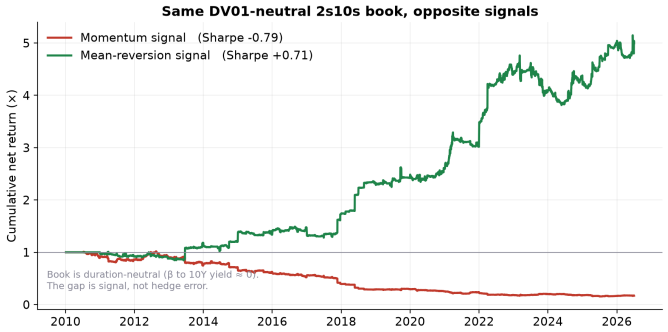


Fig. 1. The 2s10s DV01-neutral book under two signals. The book is identical; only the signal differs. Momentum loses (net Sharpe -0.79); mean-reversion earns (net Sharpe 0.71). The book's P&L has near-zero sensitivity to the 10-year yield, so the difference is signal, not hedge error.

TABLE I
NET-OF-COST PERFORMANCE, 2010–2026

Strategy	Sharpe	Max DD	Turnover/yr
Duration momentum (baseline)	0.68	-23%	22
2s10s mean-reversion	0.71	-20%	104
5s30s mean-reversion	0.66	-27%	68
2s5s10s butterfly	0.07	-38%	87

IV. RESULTS

A. The level–spread asymmetry

The paper's central result is summarized in Fig. 1. The 2s10s book is constructed identically in both lines—long the 10-year leg, short a DV01-scaled 2-year leg—and differs only in the signal that sizes it. Driven by momentum, the book returns a net Sharpe of -0.79; driven by mean-reversion, it returns 0.71. Because the book is empirically duration-neutral, this gap is not a hedging artifact: it reflects the curve's tendency to revert rather than trend. Applying the outright-momentum recipe to the curve is, over this sample, a systematic way to lose money.

B. Strategy performance

Table I reports the four pre-specified strategies with the signal appropriate to each exposure, and Fig. 2 shows their net cumulative performance. Outright duration momentum and the two curve mean-reversion trades cluster in a Sharpe range of 0.66 to 0.71 with drawdowns of roughly 20–27%. The 2s5s10s butterfly is weak (Sharpe 0.07) and carries the largest drawdown; on the evidence here it does not earn its place.

C. Does the curve add to the baseline?

Relative to the outright-momentum baseline, the HAC Diebold–Mariano statistics for the 2s10s and 5s30s mean-reversion trades are positive but small ($\mathrm{DM} = 0.83$, $p = 0.41$ and $\mathrm{DM} = 0.68$, $p = 0.49$), corresponding to an annualized return edge of about 2.9% and 2.7% that is not statistically distinguishable from zero; the butterfly is negative ($\mathrm{DM} = -1.53$, $p = 0.13$). We therefore do not claim that the curve trades outperform the baseline: they are comparable but not

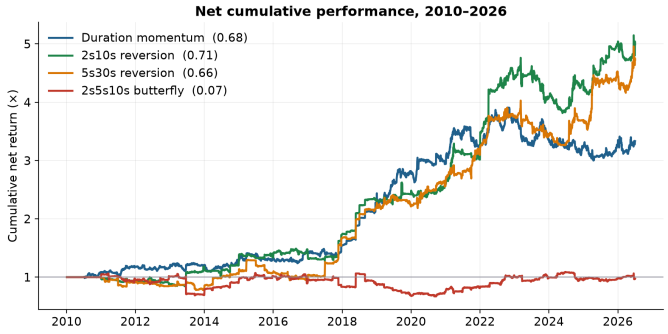


Fig. 2. Net cumulative performance of the four strategies, volatility-scaled and after transaction costs. The three viable strategies advance broadly together; the butterfly is flat to negative.

statistically distinguishable from it under this test. The honest reading is that outright duration momentum is the workhorse, curve mean-reversion is a legitimate and roughly co-equal source of return, and the butterfly is not supported.

D. Macro regime overlay

As a control and as an exportable input to a separate FX-volatility project, we classify each day into a coarse regime from the MOVE z-score and the level and trailing change of the 2s10s slope, yielding a stress state (771 days), a steepening state (1,454 days), and a flattening state (1,948 days). We report this as a descriptive conditioning variable; extending it into a real-time signal would require out-of-sample validation of its own.

V. Discussion and Limitations

Two caveats bound the strength of these conclusions. *First*, the decision to use mean-reversion rather than momentum for the curve is economically motivated but was confirmed on the full sample; a single full-sample sign is not proof, and a walk-forward, out-of-sample test is required before the curve results are treated as a hard performance claim. *Second*, the study is single-country and single-currency; the cost model, while contract-specific, is a flat tick-per-side approximation that omits market impact and financing. Neither is expected to reverse the qualitative level-versus-spread finding, but the out-of-sample validation of the curve-signal direction is the dominant open item and is what would move the curve results from suggestive to established.

VI. Conclusion

On U.S. Treasury futures over 2010–2026, outright duration trends and the yield curve mean-reverts. A momentum overlay that is correct for the former is actively wrong for the latter: an identically-hedged, duration-neutral 2s10s book loses at a Sharpe of -0.79 under momentum and earns 0.71 under mean-reversion. Matching each signal to its exposure yields three viable strategies near a Sharpe of 0.7; the curve trades are comparable to, but do not dominate, the outright-momentum baseline, and a butterfly is not supported. The result underscores that in fixed income the construction details—roll reconstruction, DV01 hedging, and signal–exposure matching—are not incidental but decisive.

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